(22) Relative Speeds on circular motions:-

4) When two objects moving around a circular path with length 'L' mtrs with speeds a & b m/s in the same direction (or) opposite direction.

(i) They meet each other at any point on the track after $\frac{L}{(a-b)}$ seconds. [when moving in same direction]

(ii) They meet each other at exactly @ the starting point after 'LCM $\left(\frac{L}{a}, \frac{L}{b}\right)$ seconds in ~~opp direction~~ same direction

(iii) They meet each other at any point on the track in after $\left[\frac{L}{a+b}\right]$ seconds when moving in opp direction.

(iv) they meet each other at exactly @ the starting point after LCM $\left(\frac{L}{a}, \frac{L}{b}\right)$ seconds in opposite direction

(v) When 3 objects moving around a circular path of length 'L' with speeds a, b & c m/s in the same direction or opposite direction.

They meet each other at any point for the first time on the track after $\left[\text{LCM}\left(\frac{L}{a-b}, \frac{L}{b-c}\right)\right]$ seconds.

(v) They meet for the first time at the starting point on the track after

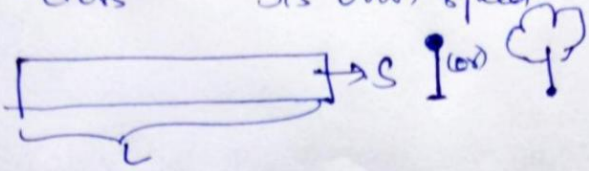
$\text{LCM}\left(\frac{L}{a}, \frac{L}{b}, \frac{L}{c}\right)$ seconds.

(63)

Trains :-

↳ A train of length (L) with speed (S) can cross a negligible length ($\cong 0$) stationary object in

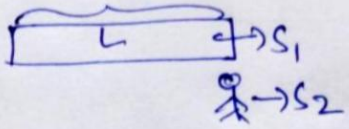
$$t_{cross} = \frac{\text{Its own length}}{\text{Its own speed}} = \frac{L}{S}$$



[negligible length stationary object means post (or) tree]

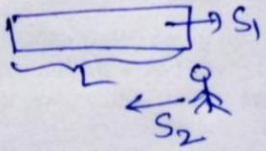
↳ A train of L & S_1 can cross a negligible length moving object (man in running / bike ..) with speed S_2 in

(i) Same direction :-



$$t_{cross} = \frac{L}{S_1 - S_2}$$

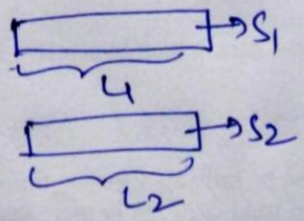
(ii) In opposite direction :-



$$t_{meet} = \frac{L}{S_1 + S_2}$$

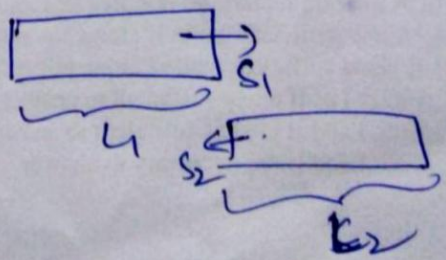
↳ two trains of lengths L_1 & L_2 and speeds S_1 & S_2 can cross / meet each other in.

(i) Same direction :-



$$t_{Cross} = \frac{L_1 + L_2}{S_1 - S_2 / S_2 - S_1}$$

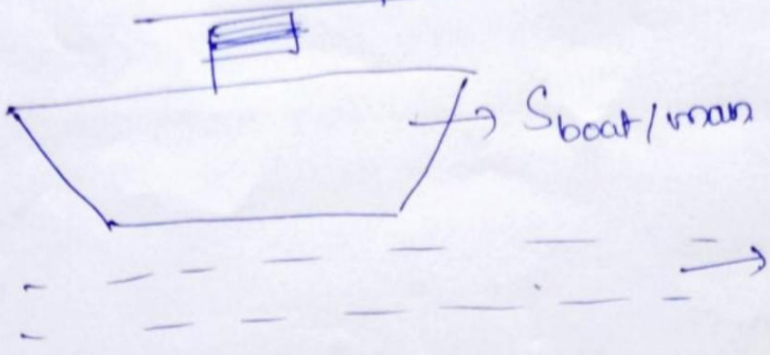
(ii) In opposite direction :-



$$t_{meet} = \frac{L_1 + L_2}{S_1 + S_2}$$

(84)

Boats & Streams.



for current.

↳ $Speed_{w/s/r}$ is speed of water/sea/river/current when there is no momentum of objects on the water.

↳ $Speed_{boat/man}$ is moving object speed on water.

↳ always moving object speed $S_{b/m} > S_{w/s/r}$ to move

↳ Upstream Speed (S_U):- when Boat direction & water direction are on opposite. against to each other

$$S_U = S_b - S_w$$

↳ Downstream Speed:- (S_d):- Directions of boat & water on same. & support to each other (+ve)

$$S_D = S_b + S_w$$

↳ S_b & S_w when we have S_U & S_D are

$$S_b = \frac{S_D + S_U}{2}$$

$$S_w = \frac{S_D - S_U}{2}$$